

CSE475: Clustering and Outlier Detection In-Class Activity

Objective

This activity will help you:

- Understand and apply clustering techniques, including **K-Means**, **DBSCAN**, and **Hierarchical Clustering**.
 - Use **DBSCAN** to detect and remove outliers, then reapply **K-Means** for cleaner clustering.
 - Compare how clustering methods work on **two datasets**:
 - **Wholesale Customers Dataset**: Spending data across product categories.
 - **Mall Customers Dataset**: Demographic and behavioral patterns (Gender, Age, Income, Spending Score).
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Datasets

- **Wholesale Customers Dataset**:
 - **Features**: Spending on Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicatessen.
 - **Focus**: Customer segmentation based on **spending behaviors**.
 - **Mall Customers Dataset**:
 - **Features**: Gender, Age, Annual Income, Spending Score.
 - **Focus**: Customer segmentation based on **demographics and spending patterns**.
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Activity Steps

Step 1: Data Preprocessing

1. Load both datasets:
 - Wholesale Customers Dataset (Wholesale_customers_data.csv).
 - Mall Customers Dataset (Mall_Customers.csv).
 2. Preprocess the data:
 - **Wholesale Customers**:
 - Exclude Channel and Region.
 - Scale the features using **StandardScaler**.
 - **Mall Customers**:
 - Encode Gender (Male=1, Female=0).
 - Scale the numerical features (Age, Income, Spending Score).
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Step 2: Outlier Detection with DBSCAN

1. Apply **DBSCAN** to detect outliers:
 - Wholesale Customers:
 - Start with `eps=1.5` and `min_samples=5`.

- Mall Customers:
 - Start with `eps=0.5` and `min_samples=5`.
- 2. Experiment with different `eps` and `min_samples` values to observe:
 - How the number of outliers changes.
 - How cluster shapes evolve.
- 3. Visualize results:
 - Use PCA to plot the clusters and highlight outliers (labeled -1).
- 4. Remove outliers:
 - Filter out data points labeled -1 by DBSCAN.

```
# Add DBSCAN labels to the dataset
data['Cluster'] = labels_dbscan
```

```
# Filter out rows where the cluster label is -1
filtered_data = data[data['Cluster'] != -1]
```

```
# Drop the 'Cluster' column if it's no longer needed
filtered_data = filtered_data.drop(columns=['Cluster'])
```

Step 3: Apply K-Means to Cleaned Data

1. Apply **K-Means** to:
 - The original dataset.
 - The cleaned dataset (after removing outliers).
2. Experiment with `n_clusters`:
 - Wholesale Customers: Try `n_clusters=3`.
 - Mall Customers: Try `n_clusters=3` and `n_clusters=5`.
3. Visualize clusters using PCA:
 - Compare how removing outliers affects cluster compactness.

Step 4: Compare with Hierarchical Clustering

1. Generate dendrograms for both datasets:
 - Wholesale Customers: Use ward linkage.
 - Mall Customers: Use average linkage.
2. Experiment with different thresholds to cut the dendrogram and assign clusters.
3. Compare the clusters formed with those from K-Means and DBSCAN.

Deliverables

- **Visualizations:**
 - PCA plots for clusters from DBSCAN, K-Means, and Hierarchical Clustering.
 - Dendrograms showing cluster hierarchies for each dataset.
 - Clear labeling, legends, and annotations to explain what the visualizations show.

- **Summary Tables:**
 - For each clustering method, provide **cluster summaries** (e.g., mean values for each feature).
 - Include a written interpretation of what the clusters represent.
 - **Reflection Report:**
 - Answer provided reflection questions.
 - Compare the effectiveness of each clustering method for each dataset.
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Reflection Questions

1. **DBSCAN:**
 - How many outliers were detected in each dataset? What do these outliers represent?
 - How did changing eps and min_samples affect the clustering results?
 2. **K-Means:**
 - How did removing outliers affect the K-Means clusters?
 - Which n_clusters value produced the most meaningful clusters?
 3. **Hierarchical Clustering:**
 - How does the dendrogram reveal relationships between clusters?
 - What happens when the threshold for cutting the dendrogram is too high or too low?
 4. **General Comparison:**
 - Which clustering method worked best for each dataset? Why?
 - How would you use these clusters in a real-world scenario (e.g., marketing)?
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Appendix: Instructional Support

Clustering Methods and Parameters

1. **DBSCAN:**
 - eps: Maximum distance for points to be in the same cluster.
 - min_samples: Minimum points required to form a dense region.
 - Output:
 - Cluster labels (-1 indicates outliers).
 - Use scatter plots to visualize clusters and outliers.
2. **K-Means:**
 - n_clusters: Number of clusters to form.
 - Output:
 - Cluster centroids and compact clusters.
 - Use PCA to visualize results.
3. **Hierarchical Clustering:**
 - Linkage methods: ward (minimize variance), average (average distance).
 - Output:
 - Dendrograms showing how clusters merge at different thresholds.

Tips for Interpretation

1. **DBSCAN:**
 - Look for dense regions and isolated points (outliers).
 - Adjust eps to balance cluster density and outlier detection.
2. **K-Means:**
 - Use the **elbow method** to select n_clusters.
 - Compare results before and after removing outliers.
3. **Hierarchical Clustering:**
 - Use dendrograms to explore sub-clusters.
 - Choose thresholds that align with meaningful splits.

Example Parameter Settings

1. **Wholesale Customers Dataset:**
 - DBSCAN: eps=1.5, min_samples=5.
 - K-Means: n_clusters=3.
 - Hierarchical: Use ward linkage.
2. **Mall Customers Dataset:**
 - DBSCAN: eps=0.5, min_samples=5.
 - K-Means: n_clusters=3 or n_clusters=5.
 - Hierarchical: Use average linkage.